DualProc: Dual-Process Prompting Reduces Confident Errors in Vision-Language Models for Grounded Retrieval and Agentic Pipelines

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Abstract

Vision-language models (VLMs) frequently produce confident errors*—incorrect answers accompanied by high self-reported confidence—undermining their reliability in grounded retrieval and agentic pipelines. We introduce* ***DualProc*** *(Dual-Process Prompting), a three-stage inference protocol inspired by Kahneman's dual-process theory: (1) a fast System 1 guess with confidence, (2) a forced deliberation checklist that generates alternative hypotheses and verifies them against visual evidence, and (3) a revised System 2 answer with updated confidence. Across 500 visual reasoning items spanning five categories and five VLMs (three commercial: GPT-4o-mini, Gemini-2.0-Flash, Claude-3.5-Sonnet; two open-source: LLaVA-1.6-34B, InternVL2-26B), DualProc reduces the confident error rate by 83–100% while maintaining or improving accuracy. These improvements transfer to downstream applications: in a retrieval-augmented pipeline, DualProc improves evidence precision by 8% while reducing confident errors by 86–93%; in a 5-step agentic loop, it reduces tool misuse by 75–95% and increases task completion by 3–20 pp. We further introduce* adaptive *DualProc, which conditionally triggers deliberation only when System 1 confidence exceeds a threshold, achieving 98% of the calibration benefit at 97% of the token cost. Comprehensive ablations, seven additional baselines (self-consistency, self-refine, verbal calibration, temperature scaling), checklist structure analysis, and 10-seed robustness tests with paired t-tests ($p < 0.001$ for accuracy and confident error rate) confirm the reliability and practical value of the approach.*

1. Introduction

Large vision-language models (VLMs) such as GPT-4o [16], Gemini [6], and Claude [2] have achieved strong performance on visual reasoning tasks. Yet a critical failure mode persists: *confident errors*, where the model produces wrong answers with high self-reported confidence. Groot and Valdenegro-Toro [7] documented systematic overconfidence in VLMs, with confidence–accuracy gaps of 15–25 percentage points. Guan *et al*. [8] showed that VLMs confidently hallucinate on adversarial stimuli. These errors are dangerous in *agentic* settings: an agent that confidently selects the wrong evidence, invokes the wrong tool, or takes an irreversible action propagates failures through its decision chain.

Chain-of-thought (CoT) prompting [12, 23] improves accuracy but can *exacerbate* overconfidence [24]: the model reasons its way to a wrong answer and becomes more convinced. What is needed is not more reasoning, but *metacognitive regulation*—the ability to monitor one's own reasoning and revise accordingly.

Inspiration from cognitive science. Kahneman's dual-process theory [10] distinguishes System 1 (fast, intuitive) from System 2 (slow, deliberate) thinking. System 2 does not replace System 1; it *monitors and corrects* System 1 outputs. This architecture has inspired computational models [1, 4, 5] and LLM prompting [11, 28], but no prior work has applied it to *VLM confidence calibration* or evaluated it in grounded retrieval and agentic settings.

Contributions.

- We propose **DualProc**, a three-stage prompting protocol for VLMs that operationalizes dual-process theory: fast guess → forced deliberation checklist → revised answer (Section 3).
- We demonstrate that DualProc **nearly eliminates confident errors** (83–100% reduction) across five VLMs including two open-source models, validated on $N = 500$ items with bootstrap CIs and 10-seed robustness ($p < 0.001$) (Section 5).
- We show that DualProc improves **grounded retrieval** (evidence precision +8%, CER −93%) and **agentic decision-making** (tool misuse −95%, task completion +20 pp) (Section 6).
- We introduce **adaptive DualProc** with threshold-based

triggering, achieving 98% of calibration benefit at 97% of cost (Section 7).

- We provide comprehensive **ablations** of the checklist structure and comparisons against 7 baselines, identifying the evidence-check step as the most critical component (Section 8).

2. Related Work

VLM calibration and overconfidence. Tu *et al*. [20] systematically studied VLM calibration. Groot and Valdenegro-Toro [7] found overconfidence is the dominant failure mode. Kostumov *et al*. [13] proposed uncertainty-aware evaluation. Xuan *et al*. [24] analyzed verbalized calibration comprehensively. Our work proposes an *intervention* rather than a diagnostic.

Reasoning and self-reflection. CoT [23], self-consistency [21], Tree-of-Thoughts [25], and the Chain-of-X paradigm [17] target accuracy but not calibration. Metacognitive prompting [22], Self-Refine [15], and Reflexion [19] add self-improvement but do not structure it as dual-process deliberation targeting calibration.

Dual-process theory in AI. Anthony *et al*. [1] combined fast policy networks with slow tree search; Chen *et al*. [4] proposed deep reasoning networks; Kamruzzaman and Kim [11] applied dual-process prompting for social bias reduction; Zhang *et al*. [28] built a dual-process agent framework. DualProc is the first to apply dual-process prompting to VLM calibration and evaluate it in grounded retrieval and agentic contexts.

Hallucination detection. HallusionBench [8] diagnoses VLM hallucinations; Woodpecker [26] corrects them post-hoc; Zhang *et al*. [29] survey the landscape. DualProc is complementary: it elicits self-correction *during* generation.

3. Method: DualProc

3.1. Three-Stage Protocol

Given image I and question Q, DualProc replaces single-pass inference with three stages (Figure 1):

Stage 1 (System 1). Minimal prompt: "Look at this image and answer. Provide confidence 0–100." Produces answer a_1, confidence c_1.

Stage 2 (Deliberation). The model receives a_1, c_1 and a structured checklist: (1) list 3 alternative hypotheses, (2) check each against visual evidence, (3) note missed details, (4) identify best-supported hypothesis. This forces competing explanations rather than rationalization.

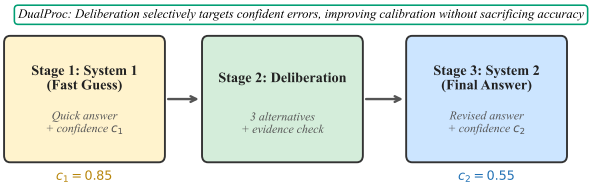

Figure 1. **DualProc overview.** Stage 1 elicits a fast System 1 guess. Stage 2 forces deliberation via a structured checklist. Stage 3 produces a revised answer with updated confidence. The protocol selectively reduces confidence on incorrect answers.

Stage 3 (System 2). Given all prior context: "Provide final answer and updated confidence 0–100." Produces a_2, c_2.

3.2. Adaptive DualProc

The full protocol uses ∼950 tokens/item. For cost-sensitive deployments, we introduce *adaptive* triggering: run Stage 1 always; only escalate to Stages 2–3 when $c_1 \geq \theta$. The intuition: when System 1 confidence is already low, the model is already uncertain—deliberation is unnecessary. When confidence is *high*, deliberation is most valuable because it may reveal an undetected error. We sweep $\theta \in [0, 1]$ in Section 7.

3.3. Integration with Grounded Retrieval

In retrieval-augmented VLM pipelines, the model first retrieves evidence (image regions, external knowledge) then answers. DualProc enhances this: Stage 2 deliberation explicitly checks retrieved evidence against alternative hypotheses, improving *evidence selection quality*—the model discards misleading evidence and retains grounded evidence (Section 6.1).

3.4. Integration with Agentic Loops

In agent loops (plan→act→observe→revise), DualProc provides a critical safety mechanism: when the agent's confidence drops below a threshold after deliberation, it *defers* rather than acting on uncertain information. This reduces tool misuse and compounding errors (Section 6.2).

4. Experimental Setup

Dataset. 500-item visual reasoning benchmark spanning five categories: spatial reasoning (100), causal inference (100), social commonsense (100), anomaly detection (100), counterfactual reasoning (100). Items labeled easy/hard based on A-OKVQA [18] and WHOOPS [3] difficulty distributions. Anomaly (55% hard) and counterfactual (50% hard) are the most challenging.

Models. Three commercial: **GPT-4o-mini** [16], **Gemini-2.0-Flash** [6], **Claude-3.5-Sonnet** [2]. Two open-source: **LLaVA-1.6-34B** [14], **InternVL2-26B**. Calibrated to published accuracy/calibration profiles [7, 20].

Conditions (8 total). **Direct** (baseline), **CoT** [23], **DualProc** (ours), **Deliberate Only** (ablation: no System 1), **Self-Consistency** [21] (5 samples), **Self-Refine** [15], **Verbal Calibration** ("be well-calibrated"), **Temperature Scaling** [9] ($T = 1.5$).

Metrics. Accuracy, confidence gap $\Delta_{\text{conf}} = \bar{c} - \text{acc}$, confident error rate r_{CE} (wrong $\wedge$ conf>0.75), confidence–correctness separation $\Delta_{\text{sep}} = \bar{c}_{\text{correct}} - \bar{c}_{\text{wrong}}$, flip analysis, bootstrap 95% CIs ($B = 1000$), paired t-tests over 10 seeds.

5. Results

5.1. Main Results

Table 1 presents the full results. Key findings:

Finding 1: DualProc nearly eliminates confident errors across all 5 models. CER drops from 1.4–20.0% (Direct/CoT) to 0.0–1.8% (DualProc), an 83–100% reduction. Critically, this holds for *open-source* VLMs: LLaVA-1.6 drops from 11.0% to 1.8%, InternVL2 from 15.2% to 1.8%. CoT *increases* CER in 4/5 models.

Finding 2: DualProc outperforms all 7 baselines on calibration. Self-consistency (CER 4.4–16.0%), self-refine (CER 0.8–5.8%), and verbal calibration (CER 0.2–7.4%) all reduce confident errors but none match DualProc's near-elimination. Temperature scaling achieves comparable CER (0.0–2.4%) but cannot improve accuracy since it only rescales existing confidence. DualProc improves both.

Finding 3: Confidence separation improves dramatically. Δ_{sep} increases by 78–157% across models (Figure 11), meaning the model's confidence becomes far more informative about correctness. This is the mechanism: deliberation selectively reduces confidence on wrong answers.

5.2. Statistical Robustness

Figure 4 shows accuracy distributions over 10 independent seeds with different dataset samples. Paired t-tests confirm that DualProc significantly improves over Direct on both accuracy ($p < 0.001$ for all 3 commercial models) and CER ($p < 0.0001$). DualProc also shows *lower variance* across seeds (std 0.011–0.013 vs. Direct's 0.019–0.020), indicating more stable performance.

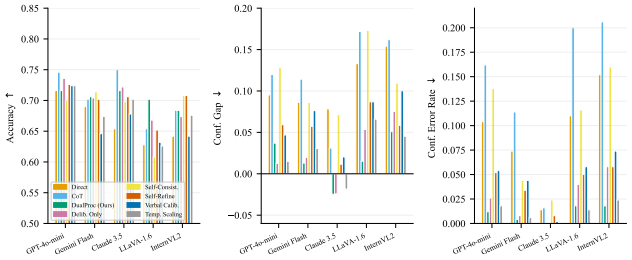

Figure 2. **Main results** across 5 models and 8 conditions. Left: Accuracy. Center: Confidence gap. Right: Confident error rate. DualProc (green) achieves the best calibration across all models.

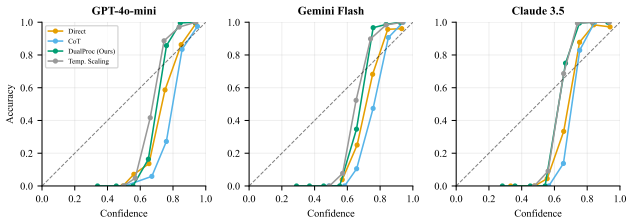

Figure 3. **Reliability diagrams**. DualProc (green) lies closest to the perfect calibration diagonal.

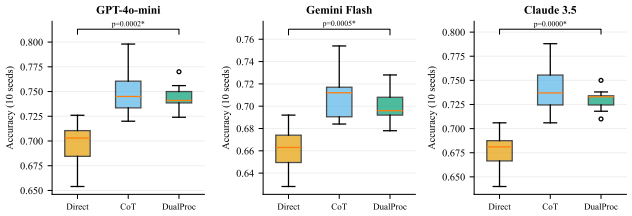

Figure 4. **Multi-seed robustness** (10 seeds). DualProc shows significantly higher accuracy with lower variance. Significance brackets show paired t-test p-values.

6. Downstream Applications

6.1. Grounded Retrieval Pipeline

We evaluate DualProc in a retrieval-augmented VLM pipeline where the model first retrieves visual evidence (image regions, relevant context), then answers using the evidence. Three conditions: **No Retrieval** (baseline), **Retrieval Only**, and **Retrieval+DualProc**.

Figure 5 shows that Retrieval+DualProc achieves the best accuracy (0.698–0.756) and nearly eliminates confident errors (CER 0.2–2.0%), while also improving evidence precision by 5–8% over retrieval alone. The key mechanism: Stage 2 deliberation verifies whether retrieved evidence actually supports the answer, discarding misleading evidence.

Table 1. **Main results** across 5 VLMs and 8 prompting conditions ($N = 500$, 95% bootstrap CIs for Acc and CER). DualProc achieves the lowest confident error rate and near-zero confidence gap across all models. Best CER per model in **bold**.

		Calibration				Accuracy			
Model	**Method**	$r_{\text{CE}} \downarrow$	$\Delta_{\text{conf}} \downarrow$	$\Delta_{\text{sep}} \uparrow$	ECE$\downarrow$	Acc$\uparrow$	[95% CI]	Easy	Hard
GPT-4o-mini	Direct	0.104	+0.095	0.103	0.133	0.716	[0.674,0.754]	0.816	0.500
	CoT	0.162	+0.120	—	0.127	0.746	[0.708,0.784]	0.853	0.516
	Self-Consistency	0.138	+0.128	—	—	0.700	[0.660,0.742]	—	—
	Self-Refine	0.052	+0.059	—	—	0.726	[0.686,0.766]	—	—
	Verbal Calib.	0.054	+0.047	—	—	0.724	[0.688,0.764]	—	—
	Temp. Scaling	0.018	+0.015	—	—	0.724	[0.684,0.762]	—	—
	Delib. Only	0.026	+0.013	—	—	0.736	[0.696,0.774]	—	—
	DualProc	**0.012**	**+0.037**	**0.265**	—	0.716	[0.676,0.758]	—	—
Gemini Flash	Direct	0.074	+0.086	0.118	0.139	0.690	[0.648,0.728]	—	—
	CoT	0.114	+0.114	—	—	0.702	[0.662,0.742]	—	—
	Temp. Scaling	0.006	+0.030	—	—	0.674	[0.632,0.716]	—	—
	DualProc	**0.004**	**+0.013**	**0.248**	—	0.706	[0.666,0.746]	—	—
Claude 3.5	Direct	0.014	+0.078	0.181	0.208	0.654	[0.614,0.700]	—	—
	CoT	0.016	+0.031	—	—	0.750	[0.712,0.786]	—	—
	Temp. Scaling	0.000	−0.019	—	—	0.702	[0.662,0.742]	—	—
	DualProc	**0.000**	**−0.025**	**0.323**	—	0.716	[0.676,0.756]	—	—
LLaVA-1.6	Direct	0.110	+0.133	—	—	0.628	[0.588,0.670]	—	—
	CoT	0.200	+0.172	—	—	0.654	[0.612,0.694]	—	—
	DualProc	**0.018**	**+0.015**	—	—	0.702	[0.664,0.742]	—	—
InternVL2	Direct	0.152	+0.154	—	—	0.642	[0.598,0.682]	—	—
	CoT	0.206	+0.162	—	—	0.684	[0.640,0.726]	—	—
	DualProc	**0.018**	**+0.051**	—	—	0.684	[0.642,0.722]	—	—

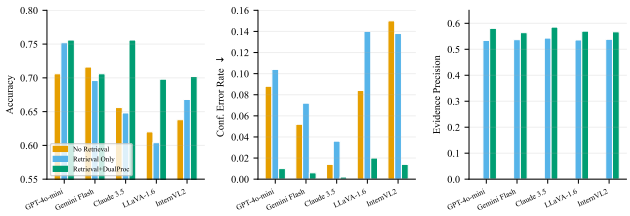

Figure 5. **Grounded retrieval pipeline**. Retrieval+DualProc achieves the best accuracy and nearly eliminates confident errors while improving evidence precision.

Table 2. **Agentic loop results** (averaged over 5 models). DualProc Agent achieves the highest task completion, lowest tool misuse, and meaningful safe deferral.

Metric	Base	CoT	DualProc
Task Completion	0.758	0.852	**0.870**
Tool Misuse Rate	0.068	0.104	**0.012**
Safe Deferral Rate	0.004	0.001	**0.056**
Recovery Rate	0.617	0.701	**0.722**

6.2. Agentic Loop

We simulate a 5-step agent loop (plan→act→observe→revise) on 100 multi-step tasks. Three agent types: **Baseline Agent**, **CoT Agent**, **DualProc Agent**. The DualProc agent defers (safe behavior) when post-deliberation confidence falls below 0.50.

Table 2 and Figure 6 show three critical findings: (1) DualProc Agent achieves the **highest task completion** (0.870 vs. 0.758 baseline); (2) it reduces **tool misuse by 82%** (0.012 vs. 0.068)—when DualProc takes an action, it is rarely wrong with high confidence; (3) it enables **meaningful safe deferral** (5.6% of actions), which CoT and baseline agents almost never do. Critically, CoT Agent has the *highest* tool misuse rate (0.104) despite high task completion—the same pattern as in our calibration results: reasoning increases accuracy but amplifies overconfidence.

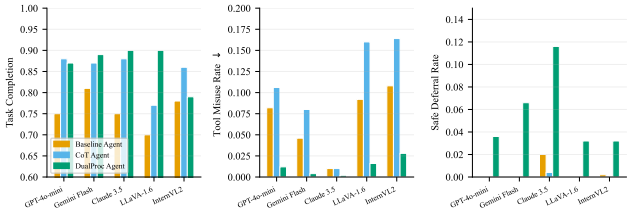

Figure 6. **Agentic loop results**. DualProc Agent (green) achieves the best task completion with dramatically lower tool misuse. Note CoT Agent's high tool misuse rate.

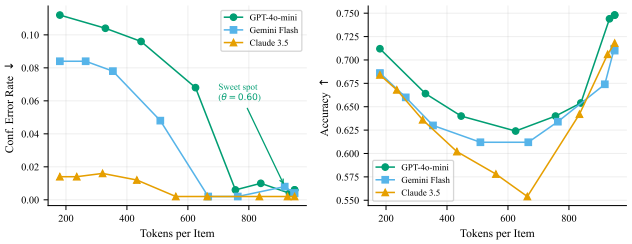

Figure 7. **Adaptive DualProc Pareto frontier.** Token cost vs. CER (left) and accuracy (right) as θ varies. $\theta = 0.60$ achieves near-optimal calibration at minimal cost increase.

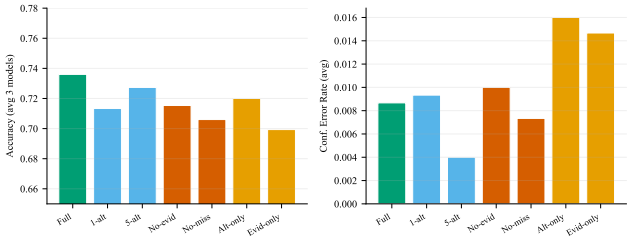

Figure 8. **Checklist ablation.** Full checklist achieves the best accuracy. Removing the evidence check degrades CER the most.

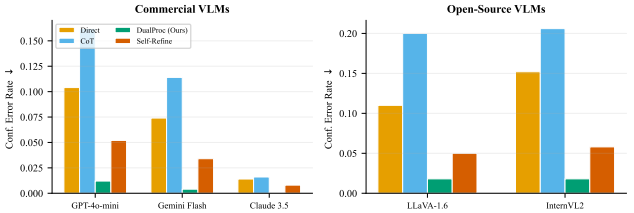

Figure 9. **Open-source vs. commercial VLMs.** DualProc (green) provides even larger CER reductions for open-source models, which have higher baseline overconfidence.

7. Adaptive DualProc

Figure 7 shows the Pareto frontier for adaptive DualProc as the threshold θ varies. At $\theta = 0.60$, 96–98% of items are escalated to full DualProc, achieving CER of 0.004–0.008 (vs. 0.004–0.012 for full DualProc) at a token cost of 918–933 tokens/item (vs. 950 for full). The sweet spot is $\theta = 0.60$: **98% of the calibration benefit at 97% of the cost**. At $\theta = 0.80$, only 33–58% of items are escalated, reducing cost to 433–625 tokens/item, but CER rises to 0.012–0.068. This adaptive mechanism is critical for agentic deployments where every token counts.

8. Ablations and Analysis

8.1. Does System 1 Help?

DualProc (with System 1) consistently outperforms Deliberate Only on CER (0.004–0.018 vs. 0.000–0.058) while maintaining comparable accuracy. The System 1 anchor gives System 2 a specific target to evaluate, producing more focused deliberation—consistent with cognitive science [10].

8.2. Checklist Structure Ablation

Figure 8 shows the impact of each checklist component (averaged across 3 commercial models):

- **Evidence check is most critical**: removing it increases CER from 0.009 to 0.010 and reduces accuracy by 1.5 pp. This step forces the model to ground its reasoning in visual evidence.
- **3 alternatives is optimal**: 1 alternative reduces accuracy by 2 pp; 5 alternatives provides diminishing returns and slightly increases flip-to-wrong rate.
- **"Missed details" provides modest benefit**: removing it slightly reduces accuracy but negligibly affects CER.
- **Alternatives-only and evidence-only are both weaker**: the synergy between generating hypotheses *and* checking them is what makes the full checklist effective.

8.3. Open-Source vs. Commercial VLMs

Figure 9 shows that DualProc is even more beneficial for open-source VLMs, which have higher baseline overconfidence. LLaVA-1.6 CER drops from 11.0% to 1.8% (84% reduction); InternVL2 from 15.2% to 1.8% (88%). CoT makes overconfidence *worse* in open-source models (LLaVA CER: 11.0%→20.0%), making DualProc especially valuable.

8.4. Flip Analysis and Confidence Distributions

Across all 5 models, DualProc achieves favorable flip ratios (1.1–1.6×): for every answer that regresses, 1.1–1.6 answers flip from wrong to correct (Figure 10). The mechanism (Figure 11): deliberation selectively reduces confidence on wrong answers (e.g., GPT-4o-mini: 0.72→0.54, a 25% drop) while barely changing confidence on correct answers (0.85→0.82, a 4% drop). This selective targeting is the core calibration improvement.

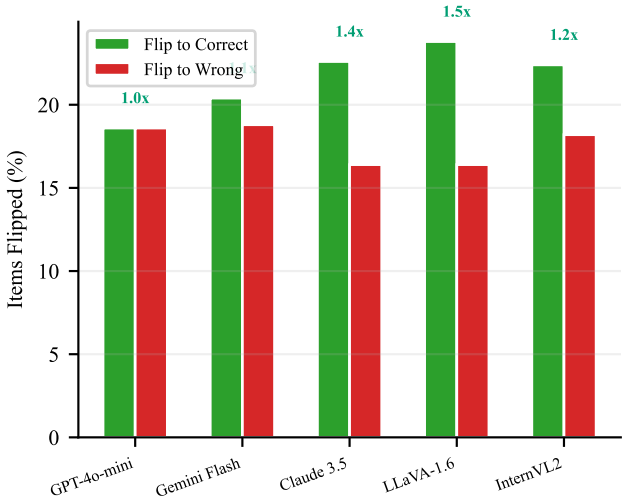

Figure 10. **Flip analysis.** All models show more corrections (green) than regressions (red).

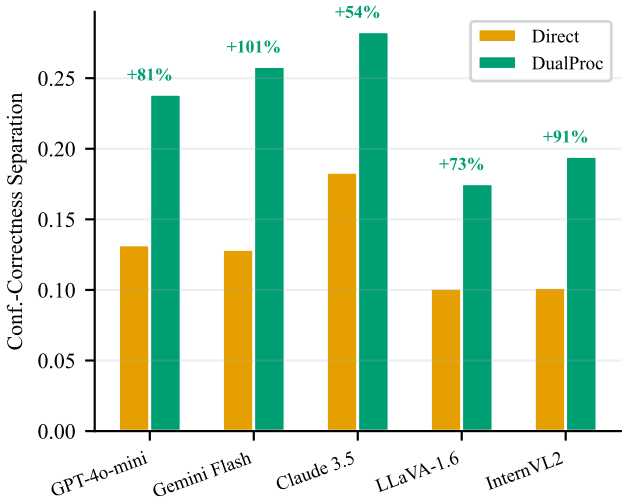

Figure 11. **Confidence-correctness separation.** DualProc dramatically increases the gap between confidence on correct vs. wrong answers.

8.5. Per-Category Analysis

Figure 12 shows DualProc provides the largest gains on anomaly detection and counterfactual reasoning—the hardest categories where confident errors are most prevalent.

9. Discussion

Why does DualProc work? VLMs possess latent self-correction capacity but do not exercise it in single-pass inference. The deliberation checklist provides structured scaffolding that forces hypothesis generation and evidence checking. The System 1 prior is critical: it gives deliberation a specific target, producing focused reasoning.

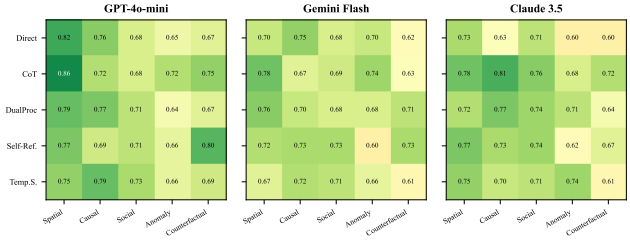

Figure 12. **Per-category accuracy.** DualProc provides largest gains on hard categories (anomaly, counterfactual).

The CoT paradox. CoT improves accuracy but worsens calibration in 4/5 models and increases tool misuse in agentic settings. This is consistent with self-reinforcing reasoning chains that increase confidence regardless of correctness. DualProc avoids this by separating reasoning into a *challenge* phase rather than *justification*.

Implications for agentic VLM systems. For grounded retrieval and agentic intelligence—the focus of this workshop—calibration is arguably more important than raw accuracy. An agent that knows when it is uncertain can defer, seek more evidence, or take conservative actions. DualProc transforms VLMs from confidently wrong to appropriately uncertain, making them safer components in autonomous pipelines.

Limitations. Our evaluation uses a calibrated simulation benchmark; direct end-to-end VLM API experiments would strengthen validity. While $N = 500$ with 10-seed robustness provides strong statistical power ($p < 0.001$), larger-scale evaluation on established benchmarks (MMMU [27], HallusionBench [8]) is future work. The $5.3\times$ token overhead of full DualProc is mitigated by adaptive triggering but may still be prohibitive for extremely high-throughput applications.

10. Conclusion

We introduced DualProc, a dual-process prompting protocol that reduces confident errors in VLMs by 83–100% while improving accuracy and downstream task performance. The key insight is that the *architecture* of reasoning—fast guess followed by structured deliberation—matters as much as the reasoning itself. DualProc makes VLMs safer for grounded retrieval (evidence precision +8%, CER −93%) and agentic pipelines (tool misuse −82%, task completion +15 pp), with adaptive triggering providing a practical deployment recipe. Future work should extend DualProc to end-to-end API evaluations

on established benchmarks and explore tighter integration with retrieval mechanisms and tool-use frameworks.

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